25901871 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCSDS100 - COMPUTER SYSTEM DESIGN & ARCHITECTURE

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Define instruction format [K] / [CO1]
2. What is a status flag? [K] / [CO2]
3. Define Flash memory. [K] / [CO3]
4. Define Input-Output Interface. [K] / [CO4]
5. What happens during the fetch cycle? [K] / [CO1]
6. What is general register organization? [K] / [CO2]
7. What is FIFO replacement policy? [K] / [CO3]
8. Which transmission is faster for short distance? [K] / [CO4]
9. Define Von Neumann Architecture [K] / [CO1]
10. What is meant by CPU operation? [U] / [CO2]

11. What is a register? [K] / [CO3]

12. Give two examples of hybrid devices. [K] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain I/O and system instructions [U] / [CO1]

14. Explain the advantages of hardwired control. [A] / [CO2]

15. Compare PROM, EPROM and EEPROM. [An] / [CO3]

16. Compare centralized and distributed arbitration. [An] / [CO4]

17. Explain instruction pipelining with suitable example. [A] / [CO2]

18. Compare paging and segmentation. [An] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19. Evaluate the impact of CPI and clock frequency on CPU performance [E] / [CO1]

20. Explain superscalar architecture with block diagram and working. [A] / [CO2]

21. Explain address structure in cache memory for direct, associative and set-associative mapping. [An] / [CO3]

22. Discuss interrupt handling mechanism step-by-step with priority schemes. [An] / [CO4]

[15x2 = 30]